Information Systems

Chapter 4

Ethical and Social Issues in Information Systems

1. What is Ethics?

Ethics means deciding what is right and wrong. In MIS, ethics means: how should individuals, companies, and governments use information systems without harming people?

> *Ethics = the principles of right and wrong that guide individuals in making choices.*

EXAMPLE

A company collects customer data from its website. Ethical use means telling customers what is collected, using it only for proper business purposes, protecting it from hackers, and not selling private data without permission. Unethical use means secretly tracking users or manipulating people.

2. Why Information Systems Create Ethical Issues

Information systems are powerful because they can store huge amounts of personal data, analyse user behaviour, track online activity, share data across networks, and copy/distribute digital content easily.

- Store huge amounts of personal data
- Analyse user behaviour
- Track online activity
- Share data across networks
- Copy and distribute digital content easily

EXAMPLE

Platforms like Facebook, Google, Daraz, or Amazon track what you search, buy, watch, your location, interests, contacts, and behaviour patterns.

3. Ethical Issues in Information Systems

Three major concerns arise when organisations collect and use customer data:

3.1 Appropriate Use of Customer Information

> *Is the company using customer data fairly and with permission?*

EXAMPLE

A clothing website saving your purchase history for recommendations is acceptable if clearly stated. Selling your phone number to advertisers without consent is unethical.

3.2 Protection of Personal Privacy

Privacy means a person's right to control their personal information. Private information includes: name, address, phone, CNIC/passport, bank details, medical records, location, messages, and online behaviour.

EXAMPLE

If a hospital system leaks patient records, it is not just a technical failure. It is an ethical and legal failure.

3.3 Protection of Intellectual Property

Intellectual property refers to creations of the human mind. Digital systems make it harder to protect because content can be copied and shared instantly.

- Software
- Books
- Music
- Videos
- Designs
- Research
- Inventions
- Business ideas
- Trademarks

EXAMPLE

Downloading paid software from an illegal website is unethical because the creator receives no payment.

4. Key Ethical Risks Raised by Information Systems

4.1 Establishing Accountability

> *Accountability means knowing who is responsible when something goes wrong.*

EXAMPLE

A bank's system wrongly deducts money. Who caused the error? The developer? The bank employee? The vendor? Without accountability, no one can be held responsible.

4.2 Setting Standards for System Quality

System quality means systems should be accurate, reliable, safe, and secure. Poor quality can harm people.

- Accuracy
- Security
- Reliability
- Backup
- Testing
- Error handling
- User safety

5. Five Moral Dimensions of the Information Age

EXAM KEY TOPIC: These five dimensions are the main ethical, social, and political issues raised by information systems.

1. Information Rights and Obligations

What rights do people have over their own information?

EXAMPLE

A user should know whether a company is collecting data, why, and how it will be used.

2. Property Rights and Obligations

How can intellectual property be protected in the digital world?

EXAMPLE

A software company's code must not be copied or sold as a pirated version.

3. Accountability and Control

Who is responsible if an information system causes damage?

EXAMPLE

If an AI hiring system rejects candidates unfairly, the company using it must be accountable.

4. System Quality

What standards should systems meet before being used?

EXAMPLE

Banking, medical, aviation, and government systems must be tested carefully because errors cause serious losses.

5. Quality of Life

How do information systems affect daily life, jobs, health, and culture?

EXAMPLE

Automation may cause job losses; employees may be over-monitored; digital divide may widen between rich and poor.

6. Basic Ethical Concepts

Responsibility

Accepting the duties, costs, and consequences of your decisions.

EXAMPLE

If a manager decides to collect customer data, the manager is responsible for protecting that data.

Accountability

Mechanisms exist to identify who took an action and who is responsible.

EXAMPLE

A system log shows which employee accessed customer data. Without logs and policies, accountability becomes weak.

Liability

Legal responsibility. If someone is harmed, the law allows them to recover damages.

EXAMPLE

If a company leaks customer bank data due to negligence, affected customers may take legal action.

7. Ethical Analysis: Step-by-Step Process

Step 1: Identify the Facts

Find out clearly what happened, who was involved, when, what data/system was involved, and who was affected.

Step 2: Identify the Ethical Issue

Ask: What is the main ethical problem? (e.g., violation of user privacy)

Step 3: Identify Stakeholders

Stakeholders include customers, employees, owners, government, society, developers, and business partners.

Step 4: Identify Options

List available choices such as stopping data sharing, asking for consent, deleting unnecessary data, or improving the privacy policy.

Step 5: Identify Consequences

Hiding the issue may lead to legal action and loss of trust. Admitting the mistake and fixing it may protect long-term reputation.

8. Ethical Principles / Rules for Decision Making

Golden Rule

Put yourself in the place of others. Do not do something to others that you would not want done to yourself.

EXAMPLE

If you would not want your private messages leaked, do not leak others' private data.

Immanuel Kant's Categorical Imperative

If everyone did this action, would it still be acceptable?

EXAMPLE

If every company secretly sold customer data, customers would lose trust in all online businesses. So the action is unethical.

Slippery Slope Rule

If an action cannot be repeated safely, it should not be done at all. A small unethical action can lead to bigger unethical actions.

EXAMPLE

A company first tracks users only for improvement, then later starts selling detailed user profiles.

Utilitarian Principle

Choose the action that creates the greatest good for the greatest number of people.

EXAMPLE

A government uses data analysis to improve traffic safety. If privacy is protected and public benefit is high, it may be ethically acceptable.

Risk Aversion Principle

Choose the action that produces the least harm or least cost. Avoid actions with high failure cost.

EXAMPLE

If a software update may cause system failure in a hospital, delay it until properly tested.

No-Free-Lunch Rule

Assume tangible and intangible objects belong to someone unless clearly stated otherwise. If something has value, someone owns it.

EXAMPLE

A paid PDF book found online is not automatically free. The author or publisher owns it.

9. Privacy in the Internet Age

Modern systems collect personal information through a wide range of channels:

- Cookies
- Web beacons
- Mobile apps
- Search engines
- Social media platforms
- E-commerce websites
- Online payment systems
- Location tracking

> *Profiling combines data from multiple sources to create detailed information about individuals. Companies use this for targeted marketing, but it creates serious privacy concerns.*

EXAMPLE

A company may know from your online activity that you are interested in fitness, travel, laptops, or medical treatment and show targeted ads. This becomes unethical when users do not know they are being tracked.

10. Intellectual Property in Information Systems

Information systems make copying very easy. Protection methods include:

Protection Type	Meaning	Example
Copyright	Protects creative work	Books, music, videos, software code
Patent	Protects inventions	New technology or process
Trade Secret	Protects confidential business knowledge	Formula, algorithm, business method

EXAMPLE

Netflix content, iTunes songs, Amazon e-books, and paid software are digital goods. Copying them without permission violates intellectual property rights.

11. System Quality and Data Quality

Bad systems create real damage. Types of problems:

- Wrong data
- Incomplete data
- Outdated data
- Software bugs
- Security holes
- Poor testing
- System crashes

EXAMPLE

If a medical diagnosis system gives wrong output due to bad data, the result can be dangerous.

12. Quality of Life Issues

Information systems affect society in both positive and negative ways.

Positive Effects	Negative Effects
Faster communication Better services Online education E-commerce Remote work Better decision making Efficient banking and healthcare	Job loss due to automation Privacy invasion Over-monitoring of employees Cybercrime Digital addiction Health issues Digital divide

13. Exam Summary: E-Commerce Ethics Scenario

A company collects customer data and secretly sells it to advertisers. Multiple ethical issues arise:

Issue	Explanation
Privacy	Customer data used without permission
Responsibility	Company made the decision
Accountability	Need to identify who approved it
Liability	Customers may take legal action
Property Rights	Customer data has value
System Quality	Security and consent systems may be weak
Quality of Life	Customers lose trust and feel monitored

14. Most Important Exam Topics

- Definition of ethics
- Ethical issues in information systems
- Five moral dimensions of the information age
- Responsibility, accountability, and liability
- Ethical analysis steps
- Ethical principles: Golden Rule, Kant's Categorical Imperative, Slippery Slope Rule, Utilitarian Principle, Risk Aversion Principle, No-Free-Lunch Rule
- Privacy and personal data protection
- Intellectual property: Copyright, Patent, Trade Secret
- System quality and data quality
- Quality of life issues

Chapter 4 explains that information systems are powerful, but they must be used responsibly because they affect privacy, property rights, accountability, system quality, and quality of life.